FOURTH AMENDMENT TO COLLABORATIVE ACCESS TEAM AGREEMENT

NORTHWESTERN UNIVERSITY ("NU"), MICHIGAN STATE UNIVERSITY ("MSU"), THE UNIVERSITY OF MICHIGAN ("UM"), WAYNE STATE UNIVERSITY ("WSU"), THE UNIVERSITY OF WISCONSIN ("UW"), and THE VAN ANDEL RESEARCH INSTITUTE ("VARI") entered into a Collaborative Access Team Agreement as of March 1, 2002 (the "CAT Agreement") for purposes of establishing a Life Sciences Collaborative Access Team ("LS-CAT" or the "CAT") for developing, facilitating, instrumenting, and operating a sector at the Advanced Photon Source at the Argonne National Laboratory. A First Amendment to Collaborative Access Team Agreement effective as of December 15, 2004 (the "First Amendment") added the University of Wisconsin ("UW") as a member of the CAT and otherwise amended the CAT Agreement. A Second Amendment to Collaborative Access Team Agreement effective as of August 1, 2005 (the "Second Amendment") added Vanderbilt University, by and through its Medical Center ("VUMC") as a member of the CAT and otherwise amended the CAT Agreement. A Third Amendment to Collaborative Access Team Agreement effective as of June 1, 2006 (the "Third Amendment") added the University of Illinois at Urbana- Champaign ("UIUC") as a member of the CAT, and otherwise amended the CAT Agreement. The purpose of this Fourth Agreement is to further amend the CAT Agreement, effective as of March 1, 2017 (the "Effective Date") and to extend the existing CAT Agreement for a period of five (5) years from the original CAT Agreement end date.

Unless indicated to the contrary, all terms defined in the CAT Agreement as previously amended shall have the same meaning in this Fourth Amendment. The terms of this Fourth Amendment shall govern in the event of any conflict with the terms of the CAT Agreement as previously amended.

In consideration of the mutual covenants and obligations set forth below, NU, UM, MSU, WSU, UW, VARI, VUMC and UIUC agree to amend the CAT Agreement as follows. Should any of the original LS-CAT members not sign this Agreement, for whatever reason, the renewal shall remain valid as between the parties provided in Section 15.

1. Subsection 11.1 is revised to state in its entirety as follows:

   "11.1 The term of the Agreement shall be for five (5) years, commencing on March 1, 2017 and ending on February 28, 2022."

2. Subsection 11.3 is added to state:

   "The absence of any of the original parties to this Agreement shall not affect the validity of any renewal or extension of this Agreement. Such renewal or extension shall be valid, as between the signatory parties."

3. Subsection 20.0 is revised to state in its entirety as follows:

   "20.0 Publication

   20.1. Each Party has the right to publish or similarly publicly disclose information arising out of research conducted solely by its employees, subject to the confidentiality obligations in Section 18.0

20.2 Each Party agrees not to unilaterally publish or similarly publicly disclose information out of research conducted a) jointly by employees of multiple Parties, b) by CAT staff, or c) jointly by CAT staff and employees of a Party or Parties, without adhering to the requirements of this provision. Publications resulting from joint research shall be joint or sole publications, as determined by the Parties conducting such research jointly.

20.3 Prior to submission for publication or public presentation of a manuscript or abstract describing the results of research conducted jointly, the publishing party will send a copy of the proposed manuscript or abstract to the non-publishing party(s). Within twenty (20) days of the other party's receipt of the manuscript, the non-publishing party(s) shall identify, in writing, for the publishing party specific information in the manuscript that is either potentially patentable or constitutes Confidential Information. Such delay shall not, however, be imposed on the filing of any student thesis or dissertation.

20.4 Upon receipt of a non-publishing party's written notice, and within the 20-day review period, the publishing party will redact any identified Confidential Information from the manuscript to be submitted or published. If a non-publishing party has identified potentially patentable information, the publishing party will delay submission or publication for a maximum of an additional sixty (60) days in order to protect the potential patentability of any invention described therein. If a non-publishing party has not identified any potentially patentable information, the publishing party may submit or publish the redacted manuscript without further delay.

20.5 If the 20-day review period expires without written notice from a non-publishing party to the publishing party, the publishing party shall be free to submit such manuscript for publication and to publish the disclosed research results in any manner consistent with professional standards.

4. Subsection 15.1 is revised to state:

"For NU: Elizabeth Adams
Executive Director
Office of Sponsored Research
1801 Maple Avenue
Evanston, IL 60201
Phone: (847) 491-4236

"For UM: Curt W. Smitka
Director of Budget and Administration
UM Office of Research
University of Michigan
4056 Fleming Building
503 Thompson Street
Ann Arbor, MI 48109-1340
Phone: 734-763-6048
Fax: 734-763-0085

"For VARI: Nathan Gras

Sponsored Research Administrator III
Van Andel Institute
333 Bostwick Ave., N.E.
Grand Rapids, Michigan 49503
Phone: (616) 234-5425